

The Spatial Dynamics of the Hawk-Dove Game

Sylvain Sylvestre Icaro Hamers

2026

Abstract

The Hawk-Dove game has been deeply studied in the mean-field approach. This project simulates the Hawk-Dove game in a 2D grid with the objective to measure the effect that the spatial feature has on the population dynamics. Applying a deterministic imitation update rule inspired by Nowak and Robert's paper *Evolutionary games and spatial chaos* we generate graphs and animations showcasing the dynamics of cooperation and competition. A phase transition is found when the payoff of the hawk on hawk interaction is 0, marking the transition into a prisoner's dilemma and generating fractal-like structures with high correlation length. The results showcase the naturally born dove strategy of agglomerating in groups while it exposes the hawks need for cooperation. These findings display the insights to be gained by simulating games with different methods.

1 Introduction

Game theory has been widely used to describe the dynamics in ecological systems, it is a useful method to model and understand natural phenomena through simple rules. These rules come in the shape of payoff matrices where the interactions between strategies are given a numerical value. In a system where agents copy the most rewarding strategy, analyzing the population of these agents through replicator equations gives insight into the dynamics of the system.

Most of the time, the replicator equations are computed with a mean-field approach, where agents have equal probability to encounter each other. However, this approach ignores the spatial aspect that we see in real-life scenarios. In order to introduce a spatial dimension to the system, researchers have implemented the game theory rules to 2D agent-based simulations. One example, and the main inspiration for this project, is the team formed by Martin A. Nowak and Robert M. May with their paper *Evolutionary games and spatial chaos* where they simulated in a grid the prisoner's dilemma. This allowed them to study the spatial dynamics of this problem, finding interesting behaviors that would be otherwise invisible.

Changing the "game" studied in Nowak and Robert's paper is as easy as changing the values of the payoff matrix. Therefore, the objective of this project is to apply these methods to the Hawk-Dove game as to observe the spatial dynamics that this ecological evolution game may display and ultimately, create pretty graphs and animations that will allow us to observe the population dynamics in real time.

2 Methods

All computer simulations were written in Julia language in the **Google Colab** environment. All the code can be found in the appendix.

2.1 Mean-Field model

The Hawk-Dove Game follows the payoff matrix seen in table 1 where the two species compete for resources. In this matrix we see the following interactions:

- **Hawk-Hawk:** In this case the two hawks fight for the same resource, getting a portion of it determined by the cost value C . The cost is always equal or greater than 0; this is not bound mathematically but rather through reason, since negative values would lead to the hawks extracting more resources than available.
- **Hawk-Dove:** In this scenario the hawk gets all the resources V and the dove flies away empty-handed.
- **Dove-Dove:** In this scenario the two doves split the resources.

	Hawk	Dove
Hawk	$(V - C)/2$	V
Dove	0	$V/2$

Table 1: Payoff matrix of the Hawk-Dove game.

In order to analytically solve this problem we are assuming all the agents in the model to have equal chance to interact. This mean-field approach leads to the general replicator equation seen in the course *Ecology and Population Dynamics*:

$$\dot{x} = x(1-x)[x(u-v) + v] \quad (1)$$

Where $u = HH - DH$ and $v = HD - DD$ (H short for hawk, D short for dove and any combination of these letters being the interaction).

In the specific case of the Hawk-Dove game this equation has the shape:

$$\dot{x} = x(1-x)\left[x\left(\frac{V-C}{2} - \frac{V}{2}\right) + \frac{V}{2}\right] \quad (2)$$

Where x represents the proportion of hawks in the system. This equation has the fixed points:

- $x = 0$: Unstable fixed point.
- $x = 1$: Unstable fixed point if $V < C$. Stable fixed point otherwise.
- $x = V/C$: Stable fixed point if $V < C$. Inexistent otherwise.

Therefore we find two main dynamics, the case in which the cost of two hawks fighting C is greater than the resource V and the reverse case. We can see these dynamics in the following diagrams:

$$(V \geq C) \text{ Hawks } \bullet \leftarrow \circ \text{ Doves}$$

$$(V < C) \text{ Hawks } \circ \rightarrow \bullet \leftarrow \circ \text{ Doves}$$

2.2 Spatial model

The model used to simulate the Hawk-Dove game in a spatial 2D grid is directly based on the model applied in [3].

The model consists of a 2D grid of size 100x100 pixels with values 0 and 1 representing doves and hawks respectively. The system evolves through "generations" as in the biological interpretation of the species having offspring in each time step. In each generation, there are two processes:

1. Each agent (pixel) is pitted against its 9 closest neighbors (including itself). Each interaction results in a payoff on table 1. All the payoffs are added together into a total payoff of the agent (Figure 1). These values are stored in a matrix matching the size of the system. This matrix is memoryless since the values reset on each generation.
2. The agents look among the 9 closest neighbors (including itself) and copies the strategy of the agent with best total payoff (Figure 2). The biological interpretation of this method is the notion that the most fitted species invades the territory of the other.

These processes are purely deterministic meaning that an initial condition will always lead to the same result. There is an article that computes the replicator equations for 2D spatial games [4]; unfortunately these equations were computed for models with non deterministic selection and didn't match the measured data in testing; no article was found describing the replicator equations for Nowak's model.

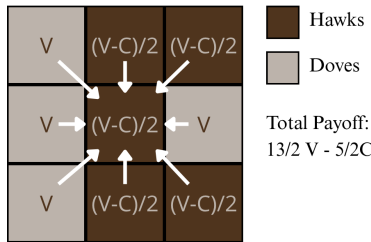


Figure 1: Diagram representing the interactions of a hawk agent with its neighbors, computing the total payoff of this hawk agent.

The self interaction is justified if we consider each pixel to be a group of hawks or doves rather than individuals. This is the interpretation used in the original paper [3].

In order to test the model we modify the payoff matrix to match the prisoner's dilemma used in Nowak's paper (Figure 3 a)) and run the same experiment, resulting in figure 3 b). As we can see, our model displays the same behaviors as in the original paper. Unfortunately, articles have the limitation of only being able to show still images; Nowak and his team

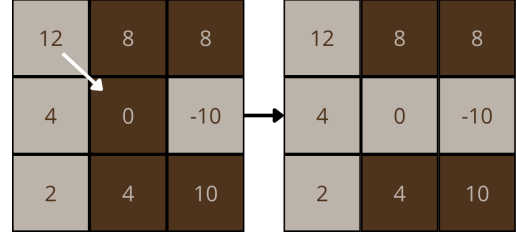


Figure 2: Diagram representing the copying method used to update the grid for the next generation.

solved this by displaying the previous generation through different colors, only showing the agents that change state; in this project we made animations of all the simulations, removing the need for these extra colors.

All the animations can be found in the appendix and are their viewing is essential to fully appreciate the dynamic geometry and fractal structures formed by this model.

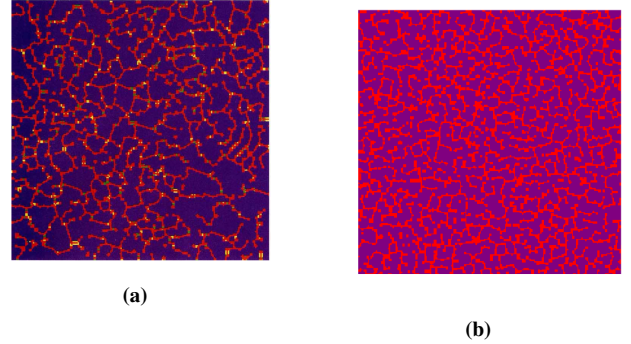


Figure 3: Figure representing the spatial simulation of the prisoner's dilemma. In a) the simulation made in the original paper [3] and in b) the spatial simulation recreated by our model using the same parameters.

2.3 Parameter analysis

The spatial aspect of the model has a big impact on the population dynamics. In order to see this impact we measured the percentage of hawks dominating the grid in relation to the cost C .

The simulations run from a value of $C = 0$ to $C = 30$ taking measures each increment of 0.5. This gives us a hawk-hawk payoff of $-10 < (V - C)/2 < 5$.

We run these measurements with an initial population of 10% hawks placed randomly around the grid. The same simulations are ran with an initial population of one hawk placed in the middle of the grid since it produces different patterns.

2.4 Correlation length

The simulations tend to exhibit fractal behavior so in order to characterize them we analyzed the autocorrelation of the grid. The reasoning is that if the patterns are truly fractal then the autocorrelation will be high. In order to compute the correlation we applied the Wiener-Khinchin theorem [2], this theorem dictates that you can obtain the correlation by

computing the reverse fourier transform of the power spectra. In order to apply this in our simulations we follow the steps:

1. Set the values of doves to -1 , otherwise these values would be invisible in the next step.
2. Apply a fast fourier transform to extract the frequency spectrum of hawks and doves.
3. Get the power spectral density by squaring the values (Figure 4).
4. Apply the reverse fourier transform to get the correlation matrix (Figure 5).
5. Normalize and extract the first row, assuming that the correlation decreases symmetrically on both axis.
6. Compute the first value that drops under $1/e$ to determine the correlation length of the system.

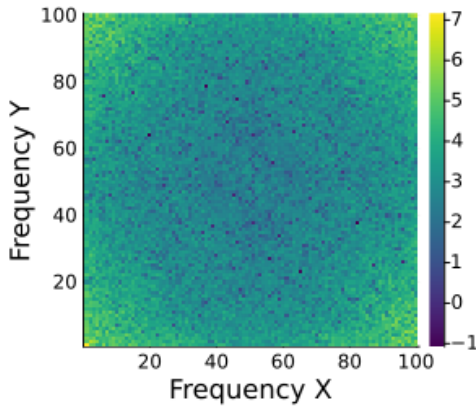


Figure 4: Power spectra of the Hawk-Dove game simulation for $(V - C)/2 = 0$. The horizontal axis measures how often the pattern repeats as you move horizontally and the vertical axis measures how often the patterns repeat as you move vertically.

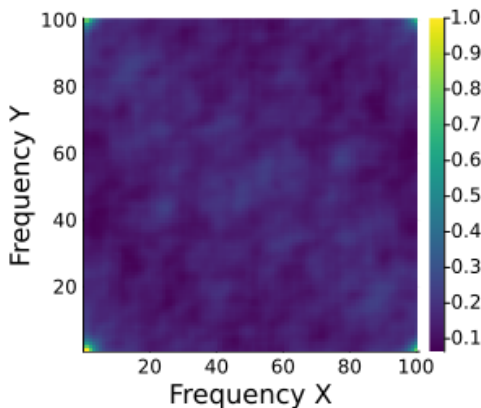


Figure 5: Correlation matrix of the Hawk-Dove game simulation for $(V - C)/2 = 0$. The axis represent the correlation of the agents as the grid is shifted in both directions.

Through these steps we measure the correlation length of each simulation, taking the same parameters as in the param-

eter analysis so as to compare both. In the cases where one of the species dominates the grid, the correlation length is infinite; these values are not interesting so we set the value of the correlation as the size of the grid for these situations.

3 Results

First, to have a reference point we display the results of the parameter analysis in figure 6 and 7. These are measured with the initial condition of 10% randomly placed hawks in the grid.

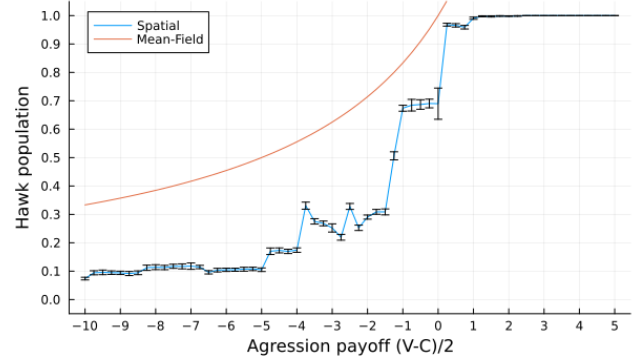


Figure 6: Graphic showcasing the percentage of hawk population in relation to the payoff of the hawk-hawk interaction. The initial condition is 10% of hawks randomly placed on the grid. The measurements are averaged over 20 runs each and the standard deviation is represented through error bars. The population predicted by the mean-field approach is also displayed as to compare the results of both methods.

There are multiple different zones in figure 6:

- $(V - C)/2 < -5$ **zone:** The cost of hawk-hawk interactions are too high, not allowing the hawks to spread in the grid. They also don't decay, so the percentage of the population is around 10% which is the initial condition.
- $-5 < (V - C)/2 < -1$ **zone:** The population reaches a balance between hawks and doves, the fixed point greatly varies depending on the specific value of the payoff. However, the standard deviation remains small indicating that this variability in values is not due to the randomness of the initial condition but rather from the specific dynamics emerging from small tweaks of the payoff.
- $-1 \leq (V - C)/2 \leq 0$ **zone:** This is a stable zone in the graph where small tweaks of the payoff doesn't affect greatly the results. However, they are more susceptible to the initial randomness.
- $(V - C)/2$ **zone:** The cost of infighting is too low, making hawks dominate the grid.

We display one graph showcasing each zone in figure 9 a), b), c) and d).

Next, in figure 8 we display the parameter analysis with the initial condition of one hawk placed in the centre of the grid. The results are not averaged over multiple runs; this is due to

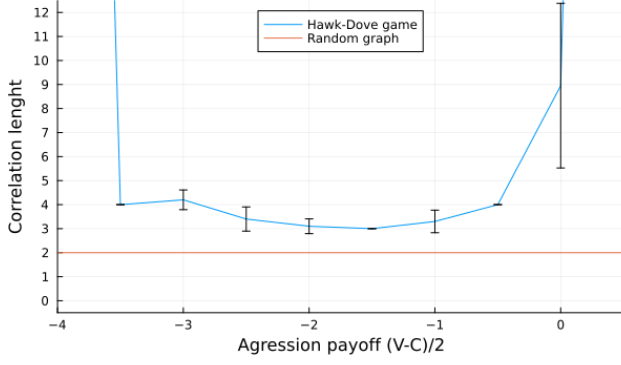


Figure 7: Graph showcasing the correlation length measured in the simulations. The initial condition is 10% of hawks randomly placed on the grid. Averaged over 20 runs and compared to the correlation length expected of a grid with a random distribution of hawks and doves. Zoomed on the values of interest; the rest of values are infinite due to the domination of one of the species.

the combination of an identical initial condition paired with the purely deterministic selection process results in identical final outcomes.

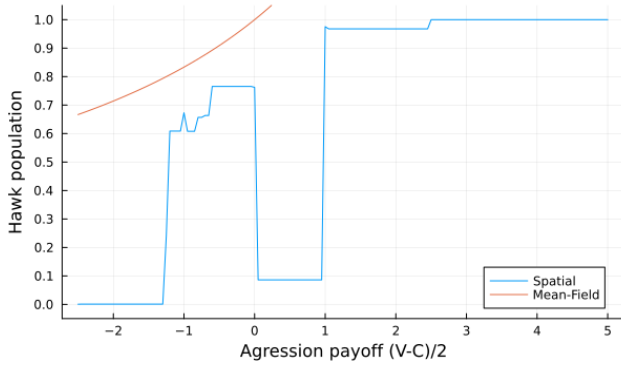


Figure 8: Graphic showcasing the percentage of hawk population in relation to the payoff of the hawk-hawk interaction. The initial condition is one hawk placed in the centre of the grid. The population predicted by the mean-field approach is also displayed as to compare the results of both methods.

The $0 < (V - C)/2 < 1$ zone is particularly interesting since the same simulations with random initial condition resulted in hawk dominance. The result for this particular case is showcased in figure 9 e). The $-1 < (V - C)/2 < 0$ zone creates particularly beautiful symmetries as showcased in figure 9 f).

4 Discussion

As observed in figure 1 the percentage of hawks simulated in a 2D grid is always lower than the predicted population by the mean-field approach. The reason, as we can see in figure 9 is the naturally emerging strategy of doves gathering in groups where they have positive feedback from each other and are safe from the hawks outside. Looking at the animations we observe that these safe groups are constantly invaded by hawks as expected from the very favored payoff; this dynamic is especially appreciated in the case of $(V - C)/2 = 0$ (Figure 9 c) where the battle of attrition

leaves both hawks without any food or injury. However, in the simulations with lower hawk-hawk payoff the doves manage to spread as quickly as they are invaded since the clusters of hawks are feeble due to infighting.

Other dynamics can be observed in different regions; in both figure 9 a) and b) we can observe how the systems reach an equilibrium state where the population of hawks flicker between two values. This is due to the high damage caused by hawk-hawk encounters, taking the system into a fixed point where the payoff of one hawk surrounded by doves is high but hawk agglomerations are heavily punished.

On the flip side we have the cases in which the hawk-hawk interactions are non negative (Figure 9 d)); as soon as this happens the system devolves into a prisoner's dilemma, where the nash equilibrium is found by filling the grid with hawks even if the final outcome is worse than filling the grid with doves. The unfortunate case scenario in which the interest of the individual ends up affecting negatively the whole.

Out of all the parameters, the null hawk-hawk payoff of $(V - C)/2 = 0$ presents the highest average correlation length (Figure 7), meaning that it also presents the highest fractal like behavior. However, the fractal feature of the model is most present in the cases of a centre hawk invading a sea of doves (Figure 10) with a correlation length of 40 pixels; the downside being that this fractal structure is temporal as the hawks eventually dominate the whole grid.

Looking at the game's dynamics when the initial condition is a single hawk in the centre (Figure 8) it can seem shocking to find a drop in hawk population when the hawk-hawk payoff is between 0 and 1; this region is supposed to be dominated by hawks. However, looking at figure 9 f), the reason becomes obvious; the balance in payoffs creates a situation where hawks can spread only in diagonals, reaching a point where it cannot keep spreading. The final result would change as soon as the initial hawk is off centre, being able to fill the grid. We can also observe from the parameter analysis that the population of hawks quickly drops when $(V - C)/2 < -1.25$, meaning that hawks cannot spread for these values and showcasing the need of cooperation between hawks to expand and prosper.

Due to the deterministic nature of the model, well chosen initial conditions can generate hypnotic symmetrical patterns as seen in figure 9 c). This behavior is very similar as the ones seen in popular cellular automaton like the *Game of life* idealized by John Horton Conway. Biology is no stranger to cellular automata, taking place in multiple natural processes like in the pattern of seashells [1]. However, the statement that hawk-dove population dynamics follow cellular automaton rules is a bold claim that I cannot make without field data to back it up.

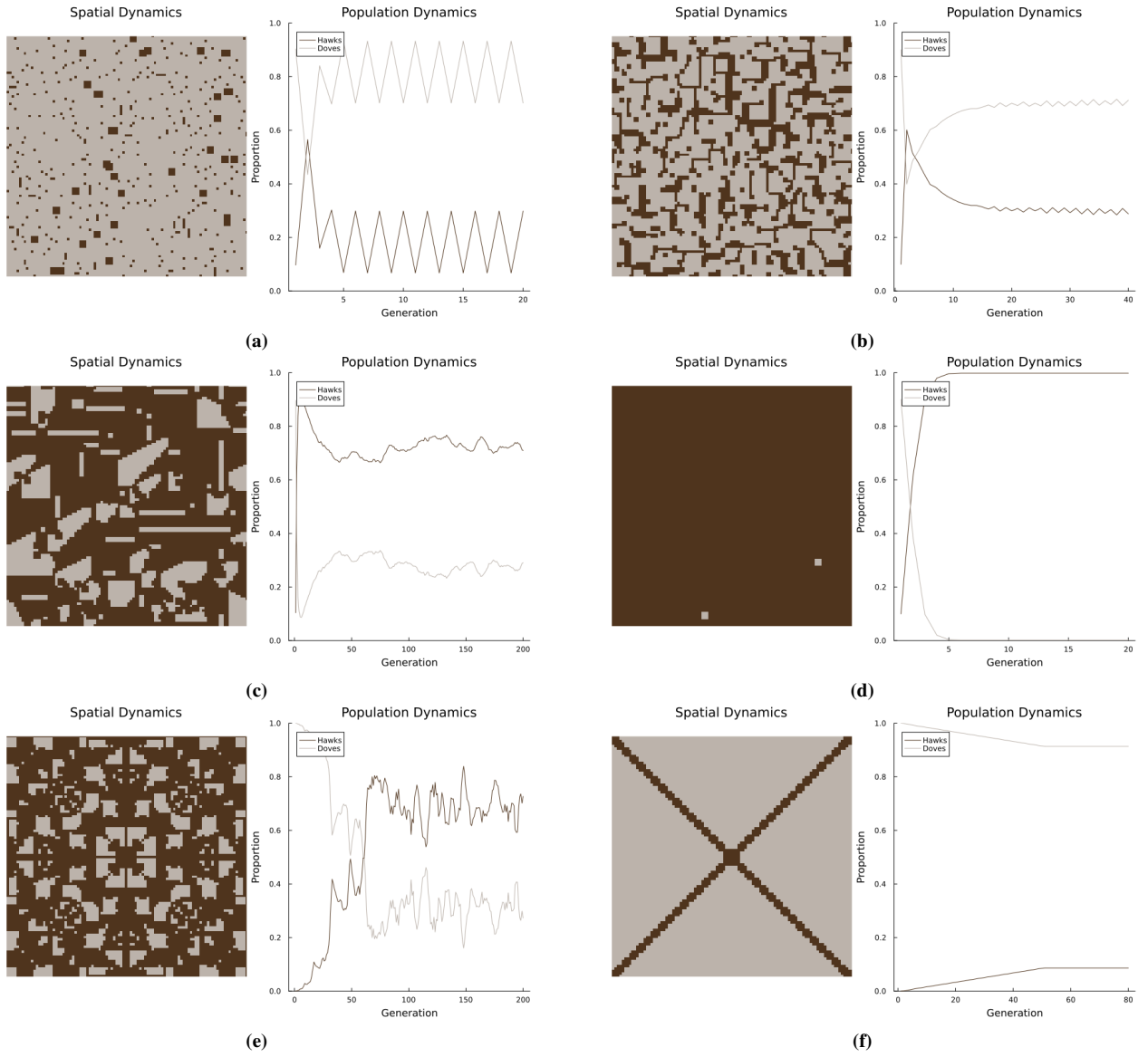


Figure 9: Graphs displaying the result of the simulations for multiple diameters with a corresponding graph showcasing the population evolution over generations. The parameters are: a) hawk-hawk payoff of -10 , ran over 20 generations; b) hawk-hawk payoff of -2 , ran over 40 generations; c) hawk-hawk payoff of 0 , ran over 200 generations; d) hawk-hawk payoff of 2 , ran over 20 generations; e) hawk-hawk payoff of -0.5 , ran over 200 generations; f) hawk-hawk payoff of 0.5 , ran over 80 generations. Graphs a), b), c) and d) were simulated with initial condition of 10% randomly placed hawks; graphs e) and f) were simulated with initial condition of one hawk in the centre of the grid.

5 Conclusions

Trough spatial simulations of the hawk-dove game we have explored the emerging dynamics and patterns that are born of this approach. From the dove strategy of agglomeration to the fractal nature of hawk invasion. These dynamics were only possible to observe in 2D grid simulations since they are a direct consequence of the spatial feature.

It is bold to claim that ecological processes perfectly follow the graphs patterns. However, from this project we are able to take away many dynamical behaviors that are present in animal interactions that were previously non-existent in the mean-field approach.

The hawk-hawk payoff value of 0 was found to be a phase transition between the hawk-dove game and the prisoner's

dilemma. Presenting high fractal nature as showcased by it's correlation length.

Further research can be made experimenting with stochastic selection methods, which would lean closer to simulations of the Ising model type or with bigger grids that would lead to smoother population dynamics.

Overall, this project resulted in a fun, dynamic analysis on the consequences of adding a spatial feature to game theory models.

References

- [1] Deborah R. Fowler, Hans Meinhardt, and Przemyslaw Prusinkiewicz. Modeling seashells. In *Proceedings of the 19th annual conference on Computer graphics and interactive techniques*, pages 379–387. ACM, 1992.
- [2] A. Khintchine. Korrelationstheorie der stationären stochastischen prozesse. *Mathematische Annalen*, 109(1):604–615, 12 1934.
- [3] Martin A. Nowak and Robert M. May. Evolutionary games and spatial chaos. *Nature*, 359(6398):826–829, 1992.
- [4] Hisashi Ohtsuki and Martin A. Nowak. The replicator equation on graphs. *Journal of Theoretical Biology*, 243(1):86–97, 2006.

Appendix

Animations

Animations are restricted by the medium of paper, the following GitHub link will guide you to a github page where the animations are fully displayed:

<https://github.com/SylvainHam/The-Spatial-Dynamics-of-the-Hawk-Dove-Game/blob/main/Animations.md>

Code

All the code written can be found in the following GitHub link:

https://github.com/SylvainHam/The-Spatial-Dynamics-of-the-Hawk-Dove-Game/blob/main/Spacial_Hawk_Dove_Game.ipynb

Extra graph

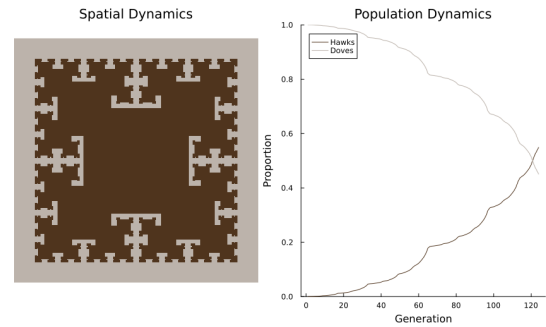


Figure 10: Graph displaying the result of the simulation of the hawk-dove game for a hawk-hawk payoff of 2, ran over 124 generations and with initial parameter of a centre hawk. Placed in the appendix due to it's early stop nature, different from the ones found in table 9.

AI statement

Google's AI Gemini tool was used to help write the code for the correlation length function and write citations in LaTeX format.